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Communication that Moves People™

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Administrator, Centers for Medicare & Medicaid Services

U.S. Department of Health and Human Services

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Thomas Keane, M.D.

Assistant Secretary for Technology Policy (ASTP)

Office of the National Coordinator for Health Information Technology (ONC)

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Submitted via www.regulations.gov

RE: Request for Information; Health Technology Ecosystem [CMS-0042-NC]

Dear Drs. Oz and Keane:

Lirio applauds CMS' commitment to using technology to transform the healthcare ecosystem. We strongly support the thesis of this request for information (RFI): "Effective and responsible adoption of technology can empower patients to make better decisions for their health and well-being."

Lirio's business model aligns directly with this vision and the Administration's goal of using AI tools to Make America Healthy Again. We appreciate the opportunity to provide our perspective as experts in the science of behavioral decision-making, AI-driven deep personalization, and the application of digital tools in healthcare.

To transform the healthcare experience and dramatically improve population health through the adoption of digital tools, the healthcare ecosystem needs to create a foundation for deeply personalized support for individual health journeys that focuses on increasing human agency. CMS and ASTP/ONC should consider building a strategy that improves health outcomes by integrating

(1) reliable clinical data with additional behavioral data that provides the broader health and social context necessary for deep personalization,

(2) personalized behavioral science to effectively leverage such data to support and guide patients based on their individual circumstances and needs, and

(3) agentic AI standards that can facilitate the orchestration of care on an individualized basis to support human agency.

Making real progress on people's health depends on pairing clinical data with digital tools that address real-life circumstances and using both to empower people to make healthy decisions both in clinical settings and at home. Allowing these components to operate together, in a seamless, secure, and patient-centered digital health infrastructure, will drive improvements in American health outcomes.

Lirio is a company that combines behavioral science with artificial intelligence to help people make positive choices about their health. Lirio uses a personalization engine and a large behavioral model to analyze users' health data and deliver tailored communications that address the root causes for people's inaction and disempowerment around their health. Using our concept of "Precision Nudging," Lirio deploys personalized interventions – with the right content, at the right time, and via the right channel – to optimize the likelihood of behavior change. We prioritize the importance of patient autonomy and empowerment, leveraging technology to allow users to meaningfully measure, understand, and influence their health and equip them with improved knowledge and agency.

Using Lirio's expertise and position as a leader in the field of digital care navigation, we offer recommendations for the following RFI questions:

- *PC-4. What features are missing from apps you use or that you are aware of today? What apps should exist but do not yet and why? What set of workflows do you believe CMS is uniquely positioned to offer?*
- *PC-5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?*
- *PC-8 b. What are specific sources, other than claims and clinical data, that would be of highest value, and why?*
- *TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?*
- *TD-4: How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?*
- *VB-2: How can key themes and technologies such as artificial intelligence, population health analytics, risk stratification, care coordination, usability, quality measurement, and patient engagement be better integrated into APM requirements?*

Our responses also touch on several other questions in the RFI such as PC-7, TD-12, and VB1.

PC-4: The Future of Digital Health (Apps and Beyond)

Most health apps today ask patients to do the heavy lifting. They ask patients to interpret charts, remember appointments, figure out food swaps, stay motivated, sync devices, and make sense of a fragmented system. What's missing are apps that act as intelligent, proactive partners in our daily lives, understanding people's motivations, adapting to their struggles, and connecting the dots between healthcare, daily life, and long-term goals.

From Lirio's perspective (and that of behaviorally aware patients) here's what's missing:

- **Just-in-time personalization**. Real-time, context-aware guidance based on individual needs and behavioral patterns
- **Emotional & motivational support**. Tools that respond not just to vitals or appointments, but to subjective experience
- **Multi-source integration**. Most apps don't unify clinical, behavioral, social, and personal data (EHR + pharmacy + nutrition + mood), forcing patients to cobble together their own 360-degree view of their

health. Ultimately, most people never achieve that holistic view because of the effort and knowledge required to get there.

- **Support for health outside of the medical office.** Solutions are often narrowly focused on clinical episodes rather than whole-person wellness and everyday life goals. Technology is not meaningfully integrated into the rhythms and contexts of everyday life.
- **Adaptive learning**. Most apps don't evolve over time. If a reminder didn't work last month, why is the app still sending it?
- **Trustworthy design**. Patients often don't trust commercial apps with their health data, and public options are limited or generic.
- **Feedback loops**. Most apps don't show the *impact* of behavior change, which is especially important for showing how everyday choices can support good health over time.

From Lirio's standpoint, and with the Cures Act as a foundation, CMS should encourage:

- **Behavior-aware digital care navigators** that understand people's unique barriers to desired health behaviors (e.g., fear of diagnosis, transportation, forgetfulness) and adapt the path forward accordingly.
- **Empathetic Digital Twin companions** that offer patients a longitudinal, virtual representation of their health journey, and evolve to reflect behavioral and health changes over time. These digital twins can be informed by data from both patient and care team to forecast, coach, and adapt in real-time.
- **Prevention-first, habit-building apps** that don't just tell people what to do but also *help them become the kind of person who does it*. These are missing because the market rewards acute, reimbursable events (e.g., prescriptions, visits), not upstream, daily improvements.
- **Health + Life coordination,** provided either via apps or via intelligence layers that can coordinate between different point solutions, that link care plans to people's schedules, grocery lists, locations, moods, etc., to infuse healthcare into daily routines. These don't exist yet because **no one owns that intersection**. CMS could be the one to convene it.

CMS is uniquely positioned to support workflows by building the **public scaffolding** for a new class of apps. This includes:

- **Standardizing access to patient-level data across payers, providers, and systems**, including behavioral, social, and engagement data — not just clinical.

- **Defining behavior-change interoperability protocols** (e.g., goals, nudges, adherence feedback) alongside FHIR and EHR standards.
- **Creating a "Trusted Health App Framework"** similar to what the VA or NHS have done — a whitelist of secure, evidence-based tools with approved access to Medicare/Medicaid data, particularly when those tools have demonstrated strong outcomes and effectiveness to improve health and reduce cost.
- **Incentivizing workflows that link benefits to behaviors**, such as earned incentives for completing nudged activities (screenings, preventive tasks).
- **Piloting AI-agent–based workflows** where behaviorally aware systems can interface with EHRs, care managers, transportation, or pharmacy services — agent-to-agent coordination on behalf of the patient.

PC-5: Encouraging Patient and Caregivers to Use Digital Health

While CMS programs center on Medicare and Medicaid, the behaviors and systems that shape health outcomes extend far beyond program eligibility. Digital engagement must support all people in building sustainable health habits across their lifespan. To foster greater adoption of digital health products, CMS must focus not only on technology access but also on behavioral engagement. Many patients and their caregivers face systemic, environmental, and motivational barriers that technology alone cannot solve. Engagement isn't solved by access to information alone; it's a behavioral design challenge. People need systems that recognize their motivations, barriers, and real-life contexts, and support their agency in choosing how, when, and through what means to engage.

A Behaviorally Intelligent Path Forward: In our response to PC-4, we detail what's missing from today's digital health ecosystem. Our recommendations on how to advance digital health will also serve to spark interest from patients and caregivers in using these tools. To meaningfully support people across the health spectrum, not just within formal care settings, CMS should envision an ecosystem where digital health products extend into everyday life. From grocery stores and fitness platforms to transportation and housing services, tools that contribute to healthier choices and improved access to care must be able to participate in a secure, interoperable, and human-centered environment.

CMS should catalyze the development and deployment of tools that:

- Personalize outreach using behavioral science — not just demographics or conditions, but individual motivations, fears, capabilities, routines, and goals — empowering people through deep, context-aware personalization.
- Leverage evidence based behavioral science frameworks and approaches to surface and resolve what stands in the way of healthy action.
- Support caregiver workflows that reinforce and align with patient-facing interventions including communication prompts, decision nudges, and wraparound support.
- Empower people and caregivers as active agents in their health, giving them meaningful choice, visibility, and control in how digital health products support their needs.
- Focus on people and health, not just patients and care — enabling proactive support in everyday life, from nutrition and movement to emotional well-being and motivation. Create space for ecosystem partners — such as grocers, transportation providers, housing services, digital health innovators like Lirio, or community organizations — to contribute to healthier living. These partners can improve outcomes not only once someone becomes a patient, but also long before, helping shift the system left toward prevention, empowerment, and agency.

On a program level, CMS can prioritize the development of such solutions in a number of ways:

- Create incentives for behaviorally personalized systems such as bonus scoring or reimbursement for solutions that drive behavior change in underserved groups.
- Support pilot programs (via CMMI) focused on population-specific behavioral interventions.
- Enable innovation within the ecosystem by establishing frameworks and protocols that allow public and private tools to interoperate flexibly with minimal friction, supporting lightweight caregiver tools alongside larger EHRs.
- Define standards for behavioral data interoperability, allowing systems to exchange goal progress, barriers encountered, and nudging success rates.
- Establish a secure, trusted, API-first federal communication infrastructure — one that allows certified public and private tools to engage individuals across diverse communication channels in a privacy-protected, standards-based ecosystem. CMS would serve not only as a payer, but as the convening digital backbone for scalable, coordinated outreach — enabling trusted partners to deliver health engagement, nudges, and education through tools of the person's choosing, from clinical apps to community resources.
- Fund public-facing digital tools or establish certification programs for public-private tools that embed behavior change strategies, allowing for flexibility while ensuring trust, compliance, and equity.

- Embed digital health education and incentives into benefit design, rewarding consistent use of digital tools that improve outcomes.
- Recognize caregivers as key participants in the behavior change ecosystem and ensure tools are designed to support and motivate them as well.

CMS has the authority and reach to normalize engagement models that are adaptive, empathetic, and behaviorally intelligent, benefiting not only patients and caregivers, but *people* in their full health and life contexts. That transformation starts not with more apps, but with smarter, more human-centered ones connected through intelligent orchestration and personalization.

PC-8 b: High Value Data Sources

Our approach leveraging behavioral science and AI seeks to use all relevant, available data to personalize messages to address individual patient contexts and barriers and subsequently drive positive health behaviors. As part of the learning process, we generate unique behavioral insights as the system matches behavioral science strategies to individual patients to drive behavioral objectives and health outcomes. In the course of our work, it has become clear that key pieces of data that can help reveal individual patient's barriers include both longitudinal behavioral records including communication interactions, and the outcomes that result from matching behavioral science techniques to patients in the context of observed outcomes.

In large populations, through this type of learning process, it becomes possible to accurately predict which users are most likely to perform various health behaviors, such as attending a cancer screening, getting a vaccine, etc. More importantly, in the course of learning across these large populations, the conditional probability that includes *how* you message that individual becomes much more useful, revealing the best methods by which to change an individual's behaviors in ways that lead to better outcomes. Thus, learnings that come from patient interactions include extremely valuable data that is difficult to obtain. However, this data is in fact obtainable, and adoption of tools that can obtain and leverage such data should be encouraged, given the large impacts this could have on population health.

Another source of similarly useful data is in the conversations that clinicians have with patients, in which they may reveal their personal barriers to performing behaviors that both the physician and patient acknowledge are important to their overall health outcomes. This type of data can be crucial in supporting patients, particularly high-risk patients whose behaviors have a great, short-term impact on their health.

TD-1: How to Stimulate Developer Interest in Innovation

CMS can accelerate developer participation by establishing a public digital infrastructure that supports scalable, behaviorally intelligent health engagement. Developers are currently held back by:

- Lack of trustable, standards-based APIs
- Fragmented messaging channels
- Limited access to timely and actionable data
- Ambiguous or burdensome regulatory environments
- Few incentives for long-term, person-centered innovation

To shift this dynamic, CMS should become a digital infrastructure convener — lowering technical barriers, aligning incentives, and enabling new forms of proactive care. Crucially, this infrastructure must serve not only Medicare beneficiaries and caregivers but support broader population health engagement to reduce long-term burden on public health resources. By enabling proactive outreach and behaviorally informed engagement earlier in life — before individuals become patients — CMS can help shift the focus of care "leftward" along the health continuum. This approach fosters better daily habits, reduces avoidable interventions, and creates a healthier, more resilient population over time. This approach must also affirm human agency — enabling individuals to shape their own health journeys through digital tools that adapt to their goals, contexts, and motivations. Deep personalization, informed by behavioral science and real-time data, is key to designing tools that do not just inform, but empower and inspire.

Short-Term Actions (Next 2 Years)

- **Launch a certified, API-first communication framework** that allows developers to plug into a standardized, privacy-preserving system for secure, real-time messaging — enabling behavioral nudges, reminders, and care coordination across native apps, RCS protocols, IoT devices, and other endpoints. This framework should also support certification of client applications (such as inbox apps, care portals, or behavioral assistant tools) that integrate with the infrastructure — ensuring ecosystem partners can tailor experiences while maintaining trust and interoperability.

- **Establish a longitudinal identity and consent framework** that allows individuals to securely authorize trusted tools and caregivers to engage on their behalf across time, settings, and life contexts. Drawing from models like Login.gov and TEFCA, this would support consistent, privacy-respecting participation in digital health services — even outside clinical encounters — while enabling continuity and trust across the ecosystem.

- **Ensure opt-in participation and clear consent pathways**, allowing CMS program participants to receive a baseline level of communication required for engagement, while enabling the broader public to voluntarily access proactive health tools through certified platforms. This helps normalize health support as part of everyday life while respecting privacy, trust, and choice.

- **Create a developer engagement program** modeled after Blue Button and Login.gov — offering support, certification, and reference implementations to ease compliance and trust-building.

• **Provide financial and reputational incentives** such as grants, scorecard credits, or tax benefits for developers building tools aligned with CMS-endorsed behavioral engagement protocols.

• **Enable tools that operate outside of formal clinical workflows**, expanding participation to community health partners, grocery and food providers, fitness platforms, and transportation services — shifting health support upstream. These tools must also respect and reinforce individual autonomy — enabling proactive choices in daily life through deeply personalized engagement, not reactive instructions.

Longer-Term Vision

• **Establish a national trust framework for health engagement APIs**, modeled after federal authentication and data exchange efforts (e.g., Login.gov, TEFCA). This framework should define secure, interoperable, and ethically governed pathways for real-time engagement — while also providing regulatory clarity that streamlines compliance rather than adding burden. Clear guardrails, consistent certification processes, and transparency around expectations will allow developers to invest confidently in tools that improve outcomes without being slowed by ambiguity or costly fragmentation.

• **Support an open ecosystem** in which certified client tools (apps, devices, interfaces) can securely access communication channels and behavioral health signals — allowing individuals to choose how they engage.

• **Define regulatory clarity and flexible interoperability standards** to de-risk long-term investment in digital health innovation.

• **Position CMS as a national convening layer**, coordinating private and public efforts in behavioral health promotion and personalized engagement.

• **Center engagement around people, not just patients** — embedding human agency into the design of digital tools and enabling dynamic, deep personalization that evolves over time.

By enabling earlier, more personalized, and context-aware engagement, CMS can reduce the need for reactive clinical care — using public infrastructure to drive better health outcomes and more efficient use of healthcare resources. This vision lays the foundation for a lifelong health ecosystem, powered by public standards and private innovation.

TD-4: CMS' Role in Establishing Publicly Available Standards-Based APIs

The future of digital health does not rest solely on opening access to patient data. It rests on opening access to all data that impacts health, and enabling intelligent systems, including agentic AI, to support human agency by coordinating the needs of individuals in context.

Today, even the limited subset of patient data that can be accessed is often constrained by paywalls, opaque licensing practices, or narrow interpretations of HIPAA that err on the side of inaction. While the law technically supports patient data access, the practical reality is that:

- Patients often cannot retrieve their full longitudinal record, even if they build or license their own software agent to do so.
- EHR systems and their API layers are typically not interoperable, not real-time, and often charge fees that deter innovation.
- Event-driven access is largely unavailable, meaning that opportunities to intervene are missed and AI tools cannot help patients at the moment when they are most able or willing to act.

Without stronger standards and enforcement, we risk an ecosystem where the most important data needed for behavior change, prevention, and empowerment remains locked behind infrastructure that was never designed to serve the patient. In such a system, even CMS's best engagement programs may fail — not for lack of intention, but for lack of actionable, timely, contextual data.

We are not advocating that all data should be made public or unconditionally accessible. But if we do not create public, enforceable protocols and governance mechanisms that respect patients' right to direct their data — across systems, in real time, and through software agents — then we will never unlock the ecosystem needed for truly personalized, proactive care.

This is not just a technical problem. It is a systems-level failing that CMS is uniquely positioned to address by:

- Establishing enforceable API access expectations, including caps on per-call pricing and timelines for fulfillment.
- Standardizing event-driven, real-time data publishing models — critical for agentic tools that rely on just-in-time intervention.
- Convening an Agentic Health Data Access Task Force, inclusive of technical, policy, and behavioral science experts, to chart a path toward intelligent, equitable data flow.

We envision a new class of public, open protocols that support:

- Agentic orchestration enabling systems, agents, and workflows to act *on behalf of* individuals, coordinating between organizations, devices, and services.

- Context-rich personalization where AI models can send and receive structured, standardized inputs (like a public version of "Model Context Protocol" or MCP) to enable transparent, auditable personalization decisions.
- Health barrier data exchange that includes not just clinical data, but data about social risks, behavioral friction, service availability, and more.

It's time to move beyond APIs that let us read records, and toward protocols that let us change lives. CMS is uniquely positioned to catalyze an open, equitable, and intelligent health technology ecosystem by defining and, more importantly, utilizing the standards that put patients and their goals at the center of interoperable action.

VB-2: Integrating Key Themes into APM Requirements

Current APM frameworks often reward retrospective outcomes while overlooking the tools that drive those outcomes — tools that anticipate needs, engage individuals meaningfully, and adapt in real time. Without incentivizing technologies that enhance human agency, personalize outreach, and close gaps in care before clinical deterioration, we dramatically limit the effectiveness of value-based care.

What CMS Can Do to Modernize APM Requirements

• **Incentivize platforms that adaptively guide and support individuals across longitudinal health journeys to improve overall outcomes**. Such platforms should support not only clinical decision-making but also individualized engagement that reflects a person's context, preferences, and readiness to act.

• **Offer reimbursement or scoring credit for AI-enabled orchestration platforms** that support services such as care coordination, transportation, digital access, and caregiver engagement. These platforms must adapt flexibly to changing needs and support whole-person care across time.

• **Incorporate more intermediate behaviors into APM metrics,** such as appointment scheduling, screening completion, or health goal setting, where there is evidence that these actions are meaningful contributors to long-term outcomes. In other words, health behaviors that have strong evidence links with positive health outcomes must be considered for inclusion in APM metrics, even if they are far upstream from the sick care that one hopes to forestall or obviate.

• **Require or reward the use of interoperable orchestration tools** that coordinate across systems, support care team collaboration, and communicate effectively with both patients and caregivers. These tools must go beyond traditional EHR integration and confront the deeper issue of data lock-in, where health information remains siloed in proprietary systems, limiting access and engagement. Ecosystem-wide interoperability should

be reframed as a shared infrastructure challenge, requiring standards, policy enforcement, and incentives to ensure data can flow securely and responsibly across care settings, organizations, and supportive services. This enables person-centered coordination to extend across the full continuum of care and life — with human agents and digital agents working together to reinforce trust, reduce administrative burden, and improve outcomes.

• **Promote real-time behavioral data as a critical input** to population health and quality measurement. Incorporating measures of patient motivation, barrier resolution, goal progress, and engagement response can offer a more comprehensive view of health system performance and equity.

• **Support solutions that enhance human agency** by enabling individuals to participate actively in their care decisions through platforms that listen, learn, and adapt. This goes beyond compliance to empower people with behavioral nudges, information, and support that are deeply personalized.

• **Embed usability and engagement criteria** into APM performance metrics to ensure that digital tools improve the lived experience of care for patients and caregivers, rather than adding friction.

• **Fund pilot projects that demonstrate longitudinal engagement**, especially in high-risk populations, using AI-powered personalization and behavioral science.

• **Allow flexibility in tool selection** while standardizing data and reporting layers, ensuring a healthy innovation ecosystem while reducing implementation complexity and cost.

APMs will only succeed at scale when they reward the technologies that make prevention, personalization, and coordination real. CMS has a unique opportunity to lead this shift — from measuring care after the fact to enabling it proactively, through smart, trusted infrastructure that supports both human and digital agents working in service of better health.

Lirio's Expertise

Lirio leverages behavioral science to guide individuals toward programs and resources that help them understand how to use their benefits and take meaningful actions in their daily lives to improve their health. Our solutions offer interventions related to self-care, preventive care, chronic care, and episodic care. Our clients have seen immediate results from a variety of interventions, such as:

- 60% of previously unengaged diabetic patients (3+ years of inactivity) completed a PCP visit after Lirio intervention
- 52% of patients achieved clinically meaningful HbA1c improvements within 6 months
- 37% of patients achieved clinically meaningful blood pressure improvements within 9 months
- 7.4x ROI within three months of diabetes intervention launch

Based on these experiences, we believe that a federally driven effort to use AI with behavior change techniques to support high-risk individuals would have dramatic effects on population health while targeting the largest cost drivers in healthcare. Therefore, we urge both CMS and ASTP/ONC to work with Lirio and similar organizations to design interventions that will be personalized, evidence-based, and sustainable.

Thank you for your consideration of Lirio's feedback and recommendations. We welcome the opportunity to share our approach and technology with you. Please contact Nancy Kavadas at nkavadas@lirio.com if you are able to meet with me and my team, or if you have any questions about our recommendations.

Regards,

Marten den Haring

Chief Executive Officer

Lirio